

**Sustained SATisfaction:
Pseudo-Boolean Optimised Indian Diet Recommendations
for Hypertension, Diabetes & PCOS**

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requirements for the degree*

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by

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CERTIFICATE

This is to certify that the entitled **Sustained SATisfaction: Pseudo-Boolean Optimised Indian Diet Recommendations for Hypertension, Diabetes & PCOS**, submitted by **Abhivyakti Srivastava** (Student ID: 1020211291) an undergraduate student of the **Department of Computer Science** has been examined. Upon recommendation by the examination committee, we hereby accord our approval of it as the presented work and submitted report fulfill the requirements for its acceptance in partial fulfillment for the degree of *DipASR in Advanced Computer Science*.

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List of Abbreviations and Symbols

NCD	Non-Communicable Diseases
PCOS	Polycystic Ovary Syndrome
BMI	Body Mass Index
BMR	Basal Metabolic Rate
GI	Glycemic Index
GL	Glycemic Load
TDEE	Total Daily Energy Expenditure
DASH	Dietary Approaches to Stop Hypertension
INDB	Indian Nutritional Databank
RDF	Resource Description Framework
OWL	Web Ontology Language
FKG	Food Knowledge Graph
ILP	Integer Linear Programming
PBO	Pseudo-Boolean Optimization
SAT	Satisfiability
CP	Constraint Programming
CP-SAT	Constraint Programming Satisfiability Solver

Chapter 1

Introduction

This thesis develops a robust and accessible dietary recommendation system tailored to Indian users, with a focus on individuals managing noncommunicable diseases (NCDs). Using an Indian food knowledge graph (FKG.in), personalized dietary constraint mapping, and SAT-based pseudo-Boolean optimization techniques, meal recommendations are developed that align with medical recommendations and typical Indian eating patterns.

Noncommunicable diseases (NCDs) such as diabetes, cardiovascular disease, and obesity have seen a sharp increase worldwide over the past few decades. NCDs killed at least **43 million** people in 2021, accounting for approximately 75% of non-pandemic-related deaths worldwide [WHO, 2024] [1]. Despite their long-term nature, NCDs are typically approached through a reactive healthcare model that prioritizes medications over prevention strategies. The current healthcare system functions primarily as a "sick care" model [2], designed to treat acute diseases and worsening symptoms rather than preventing illnesses before they develop. For chronic conditions like NCDs, this approach proves inadequate, as medications may provide temporary relief but fail to address the underlying lifestyle factors driving disease progression.

Growing evidence demonstrates that food plays a central role in both preventing and managing these chronic illnesses. Recognizing the limitations of traditional healthcare approaches, various organizations have collaborated to promote a new perspective in NCD prevention and treatment: **lifestyle medicine** [3]. The American College of Preventive Medicine defines 'lifestyle medicine' as "a scientific ap-

proach to decreasing disease risk and illness burden by utilizing lifestyle interventions such as nutrition, physical activity, stress reduction, rest, smoking cessation, and avoidance of alcohol abuse.” This approach posits that developing better lifestyle choices—which includes dietary habits—can be an effective strategy that prevents the worsening of symptoms relating to Non-Communicable Diseases.

However, adopting and sustaining dietary changes can be difficult. Many people receive advice to change their diets but struggle to follow through. In the Indian context, these challenges are amplified by socio-cultural traditions, religious norms, and affordability. Studies show that the average Indian’s diet is shaped by long-standing practices and social expectations. For instance, a diabetic patient is typically advised to avoid processed, sugary, and fried foods, yet these remain central to many daily and festive meals. As noted in a study, “there is a significant relationship between food and the social and cultural identity of a group. Hence, they are required to keep a constant watch on their daily dietary habits so as to maintain their ethical and religious food choices” [4]. In such a setting, maintaining nutritional discipline becomes even more challenging.

A variety of food recommendation systems, cooking channels, and food blogs have emerged as supplementary sources of knowledge for those making dietary changes related to NCDs. These platforms provide accessible information and inspiration, often making dietary change appear more attainable. However, they often lack cultural specificity, do not account for individual health constraints, and rarely provide long-term behavioral support. The sheer volume and inconsistency of online content can also create confusion, particularly for populations with complex dietary norms like in India.

This thesis addresses this gap by designing a culturally relevant platform for the Indian population—particularly for individuals struggling to sustain health-oriented dietary practices. The system serves as an accountability tool, supporting users in making nutritionally informed food choices without compromising the enjoyment of eating. By integrating personalized constraints with a culturally grounded knowledge base, the platform aims to bridge the gap between health-driven intent and long-term adherence.

It builds on previous work that constructs an ontology using knowledge graphs and region-specific ingredient consumption patterns. This project expands that foundation into the practical domain of personalized dietary recommendations based on health conditions and nutritional constraints.

A key strength of this work lies in its foundation: a knowledge graph developed specifically for and by the Indian community. Unlike many global food datasets, the graph is built from a corpus of over 10,000 Indian recipe websites, making it deeply representative of Indian culinary practices. This homegrown database enables the system to provide recommendations that are both nutritionally appropriate and culturally relevant.

Chapter 2

Background and Motivation

Noncommunicable diseases (NCDs) cover a range of conditions, including cardiovascular diseases, cancers, chronic respiratory diseases, diabetes. With its population of around 1.3 billion people, India contributes to more than two-thirds of all NCD-related deaths within the World Health Organization's South-East Asia Region [5]. In India, NCDs are responsible for 60% of the country's total mortality. Of these, cardiovascular diseases, chronic respiratory conditions, cancers, and diabetes are the predominant causes, contributing to nearly 82% of all NCD-related deaths.

While medications play a vital role in managing NCDs, they are not sufficient on their own. This is because medications often quickly treat the symptoms rather than attacking the root causes of chronic conditions, which are linked to poor long-term lifestyle factors such as poor diet, lack of physical activity, and stress. Sustainable management of NCDs requires consistent dietary discipline, which is difficult to achieve without structural support. It is crucial for those affected by NCDs to adhere to long-term dietary guidelines. Yet many lack access to personalized meal plans or nutritionists who can help them navigate medical advice with daily food choices. As a result, they are left to translate complex dietary requirements on their own, finally relying either on generic recommendations, or culturally misaligned habits that are difficult to sustain.

Many others abandon prescribed meal plans over time due to their restrictive nature. The desire to enjoy food remains strong, but the burden of constant decision-making—especially when balancing health, taste, and convenience—often leads to inconsistency. This challenge extends beyond patients to caregivers and nutritionists, who must regularly make informed dietary choices on behalf of others.

These concerns have informed the shape and intention of this project while offering practical, supportive tool to facilitate healthier lifestyle choices through structured, personalized dietary recommendations. This is grounded in the inherently constraint-driven nature of dietary planning which requires nutritional caretakers to choose the diet plan. Individuals often navigate multiple layers of requirements—ranging from dietary preferences and allergies to medical conditions and nutrient intake targets—when making food choices. Traditional recommender systems often fail to account for these complex, intersecting constraints in a transparent or customizable manner. Constraint satisfaction frameworks, particularly SAT-based Pseudo-Boolean Optimization, offer a structured approach to encoding such multifactorial requirements. This computational paradigm allows for clear distinctions between hard constraints (such as allergies or strict dietary exclusions) and soft constraints (such as optimal sodium intake or fiber targets), making it ideally suited for modeling nuanced dietary scenarios.

The need for such a tool was suggested by personal experience (members of my own family with diabetes and/or hypertension who struggled to adhere to dietary plans as a result of the limited variety available within their recommended diet). However, the concerns of the thesis have expanded over time to reach its current shape which focuses on three specific non-communicable diseases (NCDs): hypertension, diabetes, and polycystic ovary syndrome (PCOS). The decision to concentrate on these conditions is grounded in both their widespread prevalence and the tangible impact they have on individuals’ daily lives. Hypertension and diabetes are among the most common chronic diseases in India and globally, often progressing silently over time and contributing significantly to long-term morbidity and mortality. They represent a substantial portion of the NCD burden and, unlike acute illnesses, require continuous lifestyle management in addition to medical intervention.

Polycystic ovary syndrome (PCOS) affects an estimated 6–13% of women of reproductive age, with prevalence rates varying based on diagnostic criteria and population demographics. Its relatively low (assumed) prevalence rate compared to other NCDs sits uneasily with the observation that up to 70% of affected women remain undiagnosed worldwide [6]. PCOS significantly impacts women’s health, particularly in the reproductive and metabolic domains where it is the most common cause of anovulation and a leading cause of infertility.

The inclusion of PCOS in this study underscores a deliberate commitment to addressing health conditions that, while less prevalent, are often underrepresented in both research and healthcare technology development. Given that the management of PCOS heavily relies on sustained lifestyle modifications, particularly with respect to dietary habits, its inclusion alongside other non-communicable diseases is both meaningful and necessary. Like diabetes and cardiovascular conditions, PCOS demands long-term changes rather than quick medical fixes, naturally fitting into a system that aims at supporting personalized, health-conscious eating behaviors. By integrating PCOS into the scope of this system, the thesis not only broadens the range of addressed health conditions but also acknowledges the shared dietary challenges across these conditions. By focusing on PCOS alongside more widespread NCDs, this work seeks to foster a more inclusive approach, ensuring that the needs of many more people are considered and incorporated into tangible technology-driven interventions.

Chapter 3

Literature Survey

This thesis sits at the intersection of three key research directions in the domain of food recommendation and meal planning. It builds on the core concept of personalized, constraint-based recommendation over food knowledge graphs, as established in the *“Personalized Food Recommendation as Constrained Question Answering over a Large-scale Food Knowledge Graph”* by Chen et al.(2021) [7]. Secondly, this thesis adopts the ethos of classical constraint satisfaction problems (CSP), as detailed by Niclas Sundmark in the *“Design and implementation of a constraint satisfaction algorithm for meal planning”* [8], providing the insights into the foundation of handling complex constraint requirements. Furthermore, Karthika Subbaraj’s *“Multi-Choice Diet Recommendation Application for Indian Scenario Based on Insights from Ensemble Learning Techniques”* [9] sheds light on the geo-targeted contextualization of food recommendation systems. While built upon the principles and inspiration of these foundational works, this thesis contributes uniquely by focusing on the application of these techniques to address the critical food-related lifestyle challenges posed by Non-Communicable Diseases. Specifically designed as a regionally-contextualised scope catered to the health and food requirements of Indians, this approach is grounded in the Indian Food Knowledge Graph (FKG.In), providing the necessary structure and basis for data to manage the complex nutritional and dietary constraints associated with NCDs within the Indian food context.

3.1 Previous Work on Food Recommendation Systems

This thesis is fundamentally enabled by and builds directly upon the foundational research of the Indian Food Knowledge Graph (FKG.in) team at Ashoka University, as detailed in "*Building FKG.in: a Knowledge Graph for Indian Food*" by Gupta et al.[10]. This collaborative effort developed the comprehensive knowledge graph structure that serves as the underlying database for the present work. My involvement with the FKG.in team provided the invaluable opportunity to contribute to the critical knowledge curation phase. This included integrating thousands of recipes and their constituent ingredients sourced from various online Indian food blogging websites, a process that required ensuring efficiency and minimizing redundancy within the knowledge base. This significant prior work on expanding and refining the core FKG.in structure laid the essential groundwork for this thesis. Building upon this established and expanded knowledge base, my subsequent thesis work focuses on developing an even more nuanced representation by integrating detailed recipe nutrition information, and crucially, transforming this enriched knowledge into an applied recommendation model capable of handling complex dietary constraints, particularly those relevant to Non-Communicable Diseases within the Indian context.

To inform the design and implementation of this model, it is essential to review existing research in the domain of food recommendation systems. Researchers have previously explored different ways to tackle a similar problem for generating food recommendation systems. In their paper "*Personalized Food Recommendation as Constrained Question Answering over a Large-scale Food Knowledge Graph*" (Chen et al., 2021), the authors propose a neural knowledge base question answering (KBQA) framework for personalized food recommendation. Their work primarily focuses on the question-answer system, constraint modelling and KG augmentation. This system allows the user to input raw queries, and obtain an expanded query that incorporates all the personalised requirements. Using Natural Language Processing, the constraint modeling assigns each query word a label (positive, negative, or none) and converts these labels into embeddings that guide the neural model

in interpreting and applying dietary constraints. Here, the embedding-based constraint annotation is a technique in which various constraint types—such as positive, negative—within a user query are explicitly labeled and then transformed into vector representations (embeddings) that neural models can interpret and learn to distinguish during training. Finally, these embeddings are then processed by the KBQA module, which is responsible for retrieving recipes from the KG that satisfy the constraints mentioned in the expanded query (Chen,4).

Another research done by Niclas Sundmark called the *“Design and implementation of a constraint satisfaction algorithm for meal planning”* aimed to solve the problem of malnutrition amongst the elderly due to lack of variation in food (Sundmark, 2005,1). The author has used classical constraint satisfaction problems (CSP) to generate meal-plans. Their meal-planning system implements three main constraint satisfaction algorithms: Depth-First Branch and Bound (DFBB), Forward Checking (FC), and Partial Forward Checking (PFC). DFBB serves as the baseline algorithm, systematically exploring all possible assignments while pruning paths that exceed cost or constraint limits. Forward Checking, is a method that avoids going down the dead-end paths by preemptively skipping the future nodes that won’t satisfy the current constraint needs of the user. This makes Forward Checking a more efficient method as compared to DFBB. The third algorithm being employed is a more advanced version of Forward Checking, called the Partial Forward Checking (PFC). PFC extends FC by incorporating soft constraints, enabling the system to find near-optimal solutions when not all constraints can be strictly satisfied—a practical consideration in real-world dietary planning (Sundmark, 2005,33). Therefore, it is able to incorporate preferences, which are a softer need as compared to ‘rules’—which are optimised to have the minimum variation from the original desired outputs. While the algorithms filtered out viable recipes, this search was further optimised to provide the best results. This was done by enhancing the two models of meal-planning CSP. The first model, described as the parameter model, has variables such as time, cost, difficulty, carbohydrates, fat, energy, protein, sodium, calcium and cholesterol. These variables in the parameter model contain the set of ranges each variable can attain. In the second approach, known as the recipe model, the system treats each meal that needs to be planned as a separate variable. All of these variables share the same domain, which includes the

full list of available recipes in the database. The parameter model was optimised to provide more variation in the suggested recipes. This was done by forcefully implementing the parameter model to only allow a single recipe in common between two suggestions, forcing the algorithm to provide a relief of uniqueness in recipes that had similar variables in the parameter model. The second method, although inconclusive, was to speed up the process by checking the most liked recipes first. Lastly, the stop-and go threading method was a way that allowed the system to run in the background and start giving you partial results before it would be done checking everything. It uses threading to keep working while showing you the best results it has found so far. So even if it doesn't finish, you still get something useful.

Subbaraj (2024) in the “Multi-Choice Diet Recommendation Application for Indian Scenario Based on Insights from Ensemble Learning Techniques” further narrows the problem to build a recommendation system for those with an Indian diet. The research was done on the dataset called the ‘*MyFoodData*’ by the USDA Food central. (<https://fdc.nal.usda.gov/>) [11]. The dataset had records where the value of the feature “Indian” was either true or false. This served as the basis of the random forest classifier, which was “trained on the records where the value of the feature ‘Indian’ was False. This model was then used to predict the meals for the records where the value of the feature ‘Indian’ was True.” (Subbaraj, 2024,). The system thus adapts to the Indian diet, though it relies on a non-Indian dataset (MyFoodData) and uses feature tagging Indian == True for filtering. It made use of the close relationship between fat, proteins and carbohydrates as a system, and calories. Using the pair plot between macronutrients and the meal type, each meal was classified with a ‘meal type’. This served as a basis of assigning a user a specific meal for breakfast, lunch or dinner after their goals were calculated on the basis of variables such as height (cm.), weight (kg), gender, age and activity level– which itself was divided into five categories. These variables were used to attain the body-mass index (BMI), basal-metabolic rate (BMR) and total daily energy expenditure (TDEE) to assign a caloric restriction to each meal on the basis of either distributing the calories in an equivalent amount across all meals, or a an unequal distribution depending on the user’s preference on the weightage for a certain meal. An example being someone wanting to consume higher calories during lunch, as compared to breakfast or dinner.

3.2 Similarities

Previous research in food recommendation and meal planning shares several foundational principles that resonate with the goals of this thesis. A common theme involves the use of constraints to tailor recommendations to user requirements and context. This is central to the work of Chen et al. (2021), who framed personalized food recommendation as Constrained Question Answering (CQA) leveraging a food knowledge graph (KG) to satisfy query-based constraints. It is also fundamental to the algorithmic approach of Niclas Sundmark in meal planning, which applies the ethos of classical Constraint Satisfaction Problems (CSP). Furthermore, the utility of structured data is a shared recognition, utilized as food knowledge graphs by Chen et al. (derived from Recipe1M) and as comprehensive food databases by Subbaraj (2024) (MyFoodData). The importance of personalization and adapting recommendations to user context is also a shared objective, explored by Chen et al. through user personas and by Subbaraj for the Indian scenario using physiological metrics.

3.3 Gaps and Limitations

Despite these foundational contributions, existing research presents several gaps and limitations when addressing the specific, complex requirements of health-driven diet recommendation for Non-Communicable Diseases (NCDs) within a specific cultural context like the Indian diet. While constraints are utilized (Chen et al., Sundmark), the nature and handling of these constraints might lack the transparency needed for health-critical nutritional management. Chen et al.’s neural KBQA framework relies on embedding-based interpretations of user constraints integrated via query building models. While flexible for natural language querying, this approach may not guarantee the strict, verifiable adherence required for medically informed dietary rules and may face challenges in handling complex, overlapping, multi-condition constraints (e.g., combining requirements for multiple NCDs) with certainty. Similarly, while the ethos of classical CSP (Sundmark) provides a conceptual basis, a direct application of classical CSP solvers alone is not ideally suited for simultaneously optimizing complex, quantitative nutritional targets (like specific

grams of fat or milligrams of sodium) across multiple food items, which is essential for detailed dietary management.

A significant limitation lies in the underlying data sources and their domain specificity. The knowledge graph used by Chen et al., derived from the general Recipe1M dataset, while large, is not specifically curated for the cultural and culinary nuances of regional Indian cuisine. Similarly, despite its focus on the Indian scenario, Subbaraj (2024) relies on a non-Indian dataset—MyFoodData, primarily based on USDA Food Central. This limits the cultural and nutritional authenticity, accuracy, and applicability required for highly geo-targeted recommendations relevant to specific Indian dietary patterns and ingredients. Furthermore, the algorithmic scope and complexity handling in some systems are insufficient for detailed health management. Subbaraj’s system, for instance, restricts its machine learning approach to a narrow scope (e.g., classifying dishes into meal times based on macronutrient values) rather than holistic diet personalization or multi-objective health optimization for NCDs. The recommendation logic post-classification is often limited to basic caloric allocation, failing to account for deeper nutrient-specific goals, support multi-constraint trade-offs, or allow for cumulative day-level nutrient tracking for multiple critical nutrients beyond calories.

Finally, limitations exist in transparency and clinical utility. Systems relying on learned approximations (like some aspects of Chen et al.’s embedding interpretation) or exhibiting a black-box nature (Subbaraj) lack traceability, which is problematic in health-sensitive applications where users and professionals need to understand the rationale. The explicit, guaranteed handling of strict dietary exclusions, allergies, or specific clinical constraints is often not a primary focus.

3.4 How Does this Thesis Address These Gaps

This thesis addresses the identified gaps by synthesizing key strengths from previous research and introducing novel elements specifically tailored to the complexities of health-driven diet recommendation for NCDs in the Indian context.

Unlike approaches relying solely on learned interpretations or classical CSP, this work implements a formal constraint satisfaction model using SAT-based Pseudo-Boolean optimization (PBO). Rather than inferring constraints through embeddings or query interpretation (Chen et al.) or being limited to discrete feasibility problems (classical CSP), the developed system explicitly encodes each dietary rule—such as “sugar $\leq 10\text{g}$ ” or “no peanuts”—as a hard logical constraint solvable using techniques like Max-SAT or 0-1 Integer Linear Programming (0-1 ILP). This ensures that all dietary recommendations strictly adhere to user-defined and medically-informed limitations with verifiable rigor, crucial for a nutritional food recommendation system aimed directly at impacting the well-being of users managing health conditions. This approach excels at solving complex combinatorial constraint problems (such as those arising from multiple NCDs) efficiently and transparently, offering users traceable and explainable outputs grounded in logic rather than learned approximation. PBO is specifically chosen for its power to handle the quantitative nature and optimization aspects of complex nutritional constraints (e.g., total fat, sugar, saturated fat, sodium, and kilocalories), moving beyond the capabilities typically associated with classical CSP for this task.

Secondly, this thesis addresses the data authenticity gap by leveraging the Indian Food Knowledge Graph (FKGIn) [Gupta et al. citation, e.g., Gupta et al., 2024]. Unlike using general datasets like Recipe1M or non-domain-specific ones like MyFoodData, FKGIn is a resource meticulously built specifically for Indian cuisine from specific local sources. This provides the essential domain-specific, culturally authentic data foundation that was lacking in prior studies, ensuring the relevance and accuracy of recommendations for the target population. Furthermore, building upon the concept of structured data and the ethos of constraint satisfaction, and moving beyond simplistic recommendation logic, the developed system employs the dedicated constraint optimization techniques (PBO/ILP) discussed above to handle complex, multi-nutrient goals and trade-offs. This enables accurate cumulative tracking of multiple nutrients and the satisfaction of numerous, detailed nutritional constraints simultaneously, which is essential for managing NCDs. This goes significantly beyond approaches limited to basic caloric allocation or simple classification. This formal, logic-based approach also inherently provides transparency and explainability, addressing the black-box limitation found in some systems. It allows

for explicit encoding and handling of all types of dietary rules, including strict exclusions and clinical constraints, thereby enhancing the system's utility in health-sensitive applications and providing the necessary traceability. In essence, this thesis integrates a domain-specific knowledge base (FKGIn) with powerful constraint optimization techniques (PBO/ILP) and a focus on granular nutritional rules to deliver a transparent, accurate, and culturally relevant health-based diet recommendation system specifically designed for NCD management in the Indian scenario, thereby bridging the identified gaps in previous research.

Chapter 4

Problem Statement and Objectives

Current dietary-recommendation systems often aim to support healthy eating but still fail to adequately serve individuals managing complex health profiles—particularly within diverse cultural contexts such as India. Existing solutions, including those surveyed in the previous chapter, fall short of culturally relevant food databases (especially for Indian cuisine) and do not integrate nuanced dietary guidelines for non-communicable diseases (NCDs). They also struggle to transparently handle the complexity arising from numerous user preferences (allergies, meal timings, tastes) and strict nutritional constraints. This results in generic advice that misses users’ needs and builds the objective case for a more personalised, culturally grounded approach.

The primary objective of this thesis is to provide users with a meal-wise recipe-recommendation system that tries to implement health-aware suggestions while maintaining a large variety of choices. The proposed method combines two key processes: (i) filtering recipes according to the easy accessibility of ingredients in the Indian sub-region and (ii) offering each user a personalised “nutritional accountability partner.” The system is further refined by adding ingredient- and recipe-level components tailored to Indian cuisine—aiming to construct a detailed ontology that formally represents recipes and ingredients while also including their nutritional information (macronutrients, micronutrients, calories) and preparation methods. Such a structured representation builds an objective foundation for semantic reasoning about food choices and their implications for health.

The knowledge graph itself is constructed through ontology mapping, which connects recipes to their ingredients. One of the objectives is to further refine this

ontology by incorporating the ingredient- and recipe-level components mentioned above—formally representing recipes, ingredients, nutritional information (macronutrients, micronutrients, calories) and preparation methods. This structured representation enables semantic reasoning about food choices and their broader health implications.

The proposed system aims to integrate complex multi-disease constraints while accommodating extensive user personalisation. It formally models and incorporates specific nutritional guidelines pertinent to prevalent health conditions such as Diabetes, Hypertension and Polycystic Ovary Syndrome (PCOS). Hard constraints (e.g., allergies, dietary exclusions such as vegetarianism, meal-type requirements) are handled alongside soft constraints that optimise toward user-defined nutrient targets or ranges and individual food preferences. These disease-specific constraints are managed in tandem with general health recommendations to provide medically relevant dietary advice.

To address the core optimisation challenge, the system employs Pseudo-Boolean Optimisation (PBO) via a SAT-based solver accessed through the Google OR-Tools library. It focuses on an optimisation process that identifies optimal daily meal combinations by assigning binary values (0/1) to food items, ensuring that the defined medical, nutritional and personal constraints—expressed as linear equations—are satisfied while maximising overall satisfaction or minimising deviations from nutritional targets.

Finally, the system aims to prioritise transparency and usability in its output. Recommendations are accompanied by clear justifications detailing their nutritional composition and how they satisfy the specified constraints. Mechanisms for incorporating user feedback on suggested meals are included to refine subsequent recommendations—thereby enhancing user adherence, satisfaction and overall engagement with the dietary plan.

Chapter 5

Scope, Methodology, and Design

The food recommendation system gives the user a day-wise meal plan. A meal plan is defined as different recipe recommendations throughout the day— which includes four meal-timings i.e breakfast, lunch, snacks and dinner. The food recommendation system keeps track of the users dietary needs, preferences and the nutritional constraints allocated to them on the basis of the one, many or non-existing NCDs they might have. Therefore, while it was built keeping people with NCDs in mind, a user without any NCD can also use it.

5.1 Data Pipeline

The workflow begins with an OWL graph whose recipes are already de-duplicated via unique RecipeID identifiers. Whenever an ingredient is missing nutrient data, the system first consults a global INDB cache; failing that, it calls the API Ninjas nutrition endpoint, stores the response in the cache, and enriches the ontology so subsequent runs resolve the same item locally. If the external API returns no data, the ingredient is flagged and omitted from nutrient calculations, yet the parent recipe itself remains eligible. Hard-constraint filtering then applies only user-specific exclusions—diet type, declared allergens, and manually banned foods—without enforcing any nutrient limits. The surviving set is exported (without versioning) to eligible-recipes.json. In each user session, that file is loaded, the user’s calorie goal and NCD flags are injected, and the CP-SAT optimiser attaches nutrient-based soft-penalty variables (sodium, sugar, fibre, fat, etc.) before producing the personalised meal plan in recommended-plan.json.

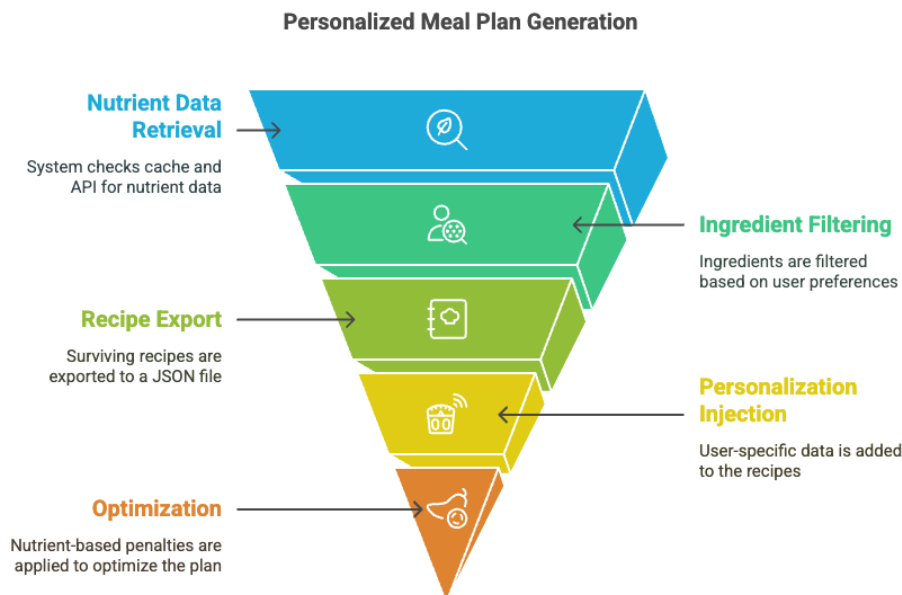


Figure 5.1: Pipeline Overview for Personalized Recipe Recommendation

5.2 Food Knowledge Graph and its Nutritional Database

The knowledge graph has different recipe properties that enhance the recipe’s meta-data. From all the variables, the ones that are of importance to the food recommendation system are the `<hasRecipeID>`, `<hasRecipeURL>`, `<hasCourse>`, `<hasDiet>`, `<hasIngredientDescription>`, `<hasActualIngredients>`. From my initial knowledge graph database, I have narrowed down the knowledge graph from over 10,000 recipes to 7177 since these 7177 provided crucial information such as the recipe’s `<hasCourse>`. The recipes are sourced from websites namely **Archana’s Kitchen**, **Masala Korb**, and **Veg Recipes of India**. However, due to the computational limitations, **my final dataset comprises 1021 recipes**.

This was the base knowledge graph— however it did not have any properties that represented the recipe’s nutritional values. In order to calculate the recipe’s nutrition, it was necessary to first find the ingredient-wise nutritional database. The two important sources that were referred to for curating the ingredient wise nutritional properties were the Indian Nutritional Database [12] and the Nutrition API offered by API Ninjas. [13]

5.3 Recipe Wise Nutrition Calculation

It should be noted that both INDB and API Ninja gave each ingredient’s nutritional value for each ingredient per 100 grams. Therefore, once the ingredients were populated, the next step was to populate the overall nutrition of the recipes. Each recipe in the knowledge graph has been appended with an ingredient description that holds information on the estimated weight of each ingredient being used. This served as a basis of calculating the recipe’s nutritional value. For each ingredient in the recipes, the nutritional value per ingredient is added and mapped to the nutritional value of the recipe. To calculate the nutritional value of a recipe, the nutritional values of each individual ingredient were aggregated. Specifically, for each nutrient, the corresponding values from all ingredients were summed to obtain the total nutrient content of the recipe. For example, if Ingredient A and Ingredient B contain 5 milligrams and 10 milligrams of sodium respectively, the resulting sodium content for the recipe would be calculated as 15 milligrams.

However, it is important to acknowledge that this method introduces a significant limitation. The direct summation of nutrient values does not accurately reflect the true nutritional profile of the final prepared dish. Nutrient retention varies substantially depending on both the type of ingredient and the cooking method employed. Certain cooking techniques, such as frying or boiling, can result in considerable nutrient loss, thereby altering the nutritional composition. Thus, a more accurate estimation would require accounting for nutrient retention factors, which vary depending on both the ingredient and the method of preparation. Unfortunately, detailed nutrient retention data is highly specific and often clinically derived, making it difficult to obtain or generalize across diverse recipes without access to specialized studies or datasets. Consequently, while the aggregation approach provides a preliminary estimate, it does not fully capture the dynamic changes in nutrient content that occur during cooking. Out of the 1021 recipes, 967 are processed with a recipe nutrition card. This is because 54 recipes did not have ingredient information, and were skipped.

5.4 Building the User Model

A key component of this system is its capacity for personalization through structured user modeling. The effectiveness of dietary recommendations is dependent on getting the individual-specific data that reflect not only nutritional preferences but also medical conditions and lifestyle habits. To achieve this, the system collects comprehensive user information during the initial login process. This information serves as the basis for encoding hard and soft constraints, which are then processed using SAT-based Pseudo-Boolean Optimization for personalized meal planning.

The model organizes user information into the independent variables and the dependent variables. The independent variables are age, sex, height, weight, activity levels, diet, medical history of non-communicable diseases and the dependent variables are caloric requirements and ‘most active before’. Caloric Requirements: Activity, height and weight variables are used to calculate the user’s total daily energy expenditure (TDEE), which is derived from the Mifflin–St Jeor equation [14]. TDEE reflects the actual calories burned per day, including physical activity. If the user’s Body Mass Index (BMI) is ≥ 25 , indicating overweight or obesity, a reduced calorie target (typically TDEE minus 500 kcal) is suggested, subject to a safe minimum threshold. Otherwise, users may adopt their TDEE as their calorie goal. This approach personalizes energy targets while ensuring they remain both practical and health-conscious.

As suggested by Mass General Brigham [15], individuals with PCOS are advised to consume their heaviest meal during their most active part of the day to support better hormonal and metabolic regulation (Mass General Brigham, n.d.). This is used to identify the optimal timing for the more caloric meal for a person with PCOS, ensuring it aligns with the user’s peak activity period for better metabolic and hormonal outcomes.

More about these variables, and their significance in the constraint building has been described in the next section.

5.5 Constraints

The range of user modelling variables—such as dietary preferences, nutritional needs, allergies, health conditions, and lifestyle factors—creates a combinatorial problem when attempting to generate personalized dietary recommendations. Each user represents a unique combination of these variables, and when mapped against a large and diverse recipe database, the number of possible user-recipe pairings grows exponentially. This complexity is further compounded by the need to satisfy both needs and preferences. Bridging this gap requires the application of constraints that effectively filter and match recipes based on user-specific conditions. The intersection of these multidimensional user requirements with a vast but finite recipe space ultimately formulates a constraint optimization problem.

These needs and preferences are divided into two categories, hard constraints and soft constraints. The hard constraints are the constraints that must be strictly satisfied, whereas the soft constraints are those that can be violated but must incur a penalty for the same. In this model, **there are 4 hard constraints and 20 soft constraints.**

In this model, the hard constraints are built on the user’s dietary preferences variables i.e. non-vegetarian, vegetarian or vegan diet, allergies, and meal-wise exclusion and the caloric constraints for PCOS.

1. **Diet** - Absolute filtration of the recipes :

$$\text{non-vegetarian diet} \supseteq \text{vegetarian diet} \supseteq \text{vegan diet}$$

2. **Allergies** - There is a predefined list of allergies such as dairy, gluten, nuts, soy and shellfish. The list has been taken from the US Food and Drug Administration (FDA) [16]. In the case that a user’s allergy is not listed, they can input it and any recipes with the defined ingredient(s) will be filtered as well.
3. **Meal Wise Exclusions** - This is an option available to users who might want to avoid a specific type of food item or recipe preparation during the meal planning. For example, User A wants to avoid fried food or desserts at night.

4. **Meal Wise Caloric Constraints for PCOS** - Users with PCOS are advised to eat only 10% of their total daily caloric needs during dinner, and eat their heaviest meal before the most active time of the day. If a user is most active after lunch time, their lunch is allocated with the most calories in a day, i.e 40% of daily calories. The remaining calories are divided equally amongst the other three meal-times [17].

I processed the hard constraints up front that prune out all ineligible recipes, so that when I finally invoke the heavier NCD constraint routines (diabetes, PCOS, hypertension), they only run on a much smaller, already-filtered set—saving both time and computational resources. These soft constraints are dependent on variables such as the nutritional recommendations for the NCDs, Post filtration the soft constraints are processed with the following suggestions from highly credible medical publishers as well as authors

1. **Hypertension** - The DASH [18] recommended daily sodium intake for a person with hypertension is between 1500-2300 mg of sodium per 2000 kcal. It also recommends a diet rich in potassium. According to the American Hearts Association (AHA) [19], “if you are trying to prevent or treat high blood pressure, the AHA recommends 3,500 mg to 5,000 mg of potassium daily”. Furthermore, according to the evidence reviewed here, the minimum daily dietary fiber for adults with hypertension should be 28 g/day for women and 38 g/day for men, with each extra 5 g/day estimated to reduce systolic BP by 2.8 mm Hg and diastolic BP by 2.1 mm Hg. Additionally, Medline Plus recommends to keep the fat levels 27% of daily calories, apart from saturated fat being under 6% of daily calories [20].
2. **Diabetes** - Sugar should not be exceeded for more than 5% of the calorie intake, the minimum fiber intake should be 14% per 1000 kcal, sodium less than 2300 mg/2000 kcal, Limit fat to 25%-35% of your total daily calories [21]. Additionally, there should not be more than 10% of saturated fat for your total calorie intake.
3. **PCOS** - Dinner should be aimed to be capped at 10% of the user’s total daily calories, makes the user-specified “most active” slot dominate the others, and keeps total fat under 30%, saturated fat under 10%, and free sugar

under 5% the daily calorie goal. Furthermore, the AIA suggests “doctors recommend the heart-friendly DASH diet to help with hypertension treatment and that DASH could also help with glucose control and keep body fat from being distributed around the belly in women with PCOS”. A similar constraint as compared to hypertension is thus also applied to PCOS i.e sodium should be between 1500-2300 mg of sodium per 2000 kcal.

4. **Other** - To discourage trivial solutions where the solver avoids assigning any recipes due to overly restrictive constraints, a small penalty is introduced for each unselected recipe. This encourages the model to prioritize feasible assignments even when strict nutritional or medical conditions are in place.

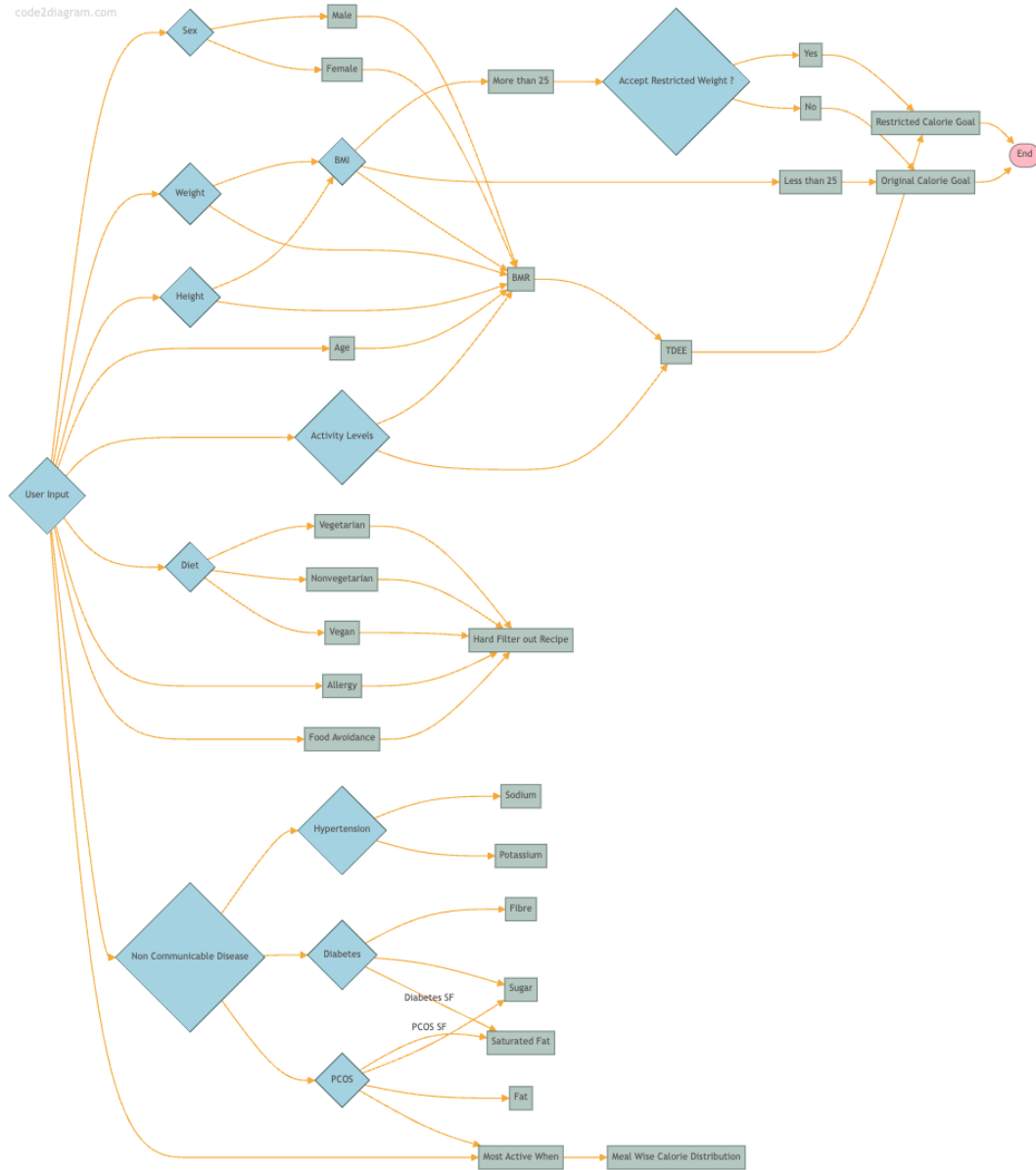


Figure 5.2: A Flowchart Depicting the Interdependency of the Constraints

5.6 Optimisation

In practical meal-planning applications, finding—or even inventing—recipes that satisfy every nutritional requirement at once is extraordinarily hard. To address this, the optimisation framework adopts a satisficing philosophy: it prefers generating a feasible meal plan that meets most goals to rejecting all options in pursuit of an unattainable ideal. The recommendation engine is built with

`ortools.sat.python.cp_model`, part of Google OR-Tools. The recommender’s code specifically utilizes the CP-SAT solver, which is a powerful component of OR-Tools designed for constraint satisfaction and optimization problems. It encodes the

meal-planning task as a combination of integer linear programme (ILP) in which hard constraints (diet type, allergy exclusions, serving quotas, meal-time slots, etc.) are modelled with integer linear equalities or inequalities, and soft constraints (e.g., preferred nutrient ranges or disease-specific limits) are converted into penalty variables that the objective function minimises. In this thesis the key components of OR-Tools utilized are:

1. **Model Creation** - The model defined as `cp_model.CpModel` is the foundation for defining the optimization problem. It allows for the creation of variables and the addition of constraints and objectives. The code uses the `cp_model.CpModel()` to create a constraint programming model and `cp_model.CpSolver()` to solve it.
2. **CP-SAT Solver**- The CP-SAT Solver is a constraint solving tool offered by OR Tools. According to the CP-SAT Solver guide, the solver *“requires you to define your optimization problem using integers only since in order to increase computational speed, the CP-SAT solver works over the integers only”* [22]. This means, since every constraint is defined as a linear expression, the CP-SAT solver’s native integer-linear-programming architecture is an ideal—and highly efficient—fit for constructing and solving the defined constraints.
3. **Constraints** – The code adds constraints to the model through `model.Add(constraint)`. This applies to both hard and soft constraints. For example, the PCOS dinner calorie cap is modeled using: `model.Add(cal_for("dinner") <= dinner_target)`.

In contrast, soft constraints allow deviation but incur penalties. For example:
`model.Add(min_s - current_sodium <= sodium_under).`

4. **Objective Function** – The optimization objective minimizes the total penalty. This is encoded as: `model.Minimize(sum(penalties))`
5. **Decision Variables** – The code defines decision variables such as `model.NewBoolVar(f"select_{i}")`, where Boolean variables are used to represent whether a specific recipe is selected or not.
6. **Solver Execution** – This is the step where the OR-Tools solver

CP-SAT return values

The CP-SAT solver returns one of the status values shown in the table below. In this example, the value returned is `OPTIMAL`.

Status	Description
<code>OPTIMAL</code>	An optimal feasible solution was found.
<code>FEASIBLE</code>	A feasible solution was found, but we don't know if it's optimal.
<code>INFEASIBLE</code>	The problem was proven infeasible.
<code>MODEL_INVALID</code>	The given <code>CpModelProto</code> didn't pass the validation step. You can get a detailed error by calling <code>ValidateCpModel(model_proto)</code> .
<code>UNKNOWN</code>	The status of the model is unknown because no solution was found (or the problem was not proven <code>INFEASIBLE</code>) before something caused the solver to stop, such as a time limit , a memory limit, or a custom limit set by the user.

Figure 5.3: CP-SAT Solver's Values, Source: Google OR Tools

(`solver.Solve`) attempts to find a solution that satisfies all hard constraints and minimizes the objective function (i.e., the total penalty). The solver can return **one of five statuses**: *optimal*, *feasible*, *infeasible*, *invalid model*, or *unknown*.

7. **Solution Interpretation** – After solving, the code checks the solver's status using `solver.StatusName`. If an optimal or feasible solution is found, it retrieves the values of the decision variables via `solver.Value(x[i])` to determine which recipes were selected. It also evaluates the penalty variables using `solver.Value(p_var)` to assess the extent of constraint violations and their contribution to the total penalty.

The following table summarizes solver statuses and their implications. In general, solutions with *optimal* or *feasible* status are desirable.

Chapter 6

Work Done – Implementations, Challenges, Mitigations

6.1 Constraint-Based Optimization Models

The development of effective food recommendation needs a system beyond generic suggestions that require tailoring recommendations to individual user needs, encompassing a complex interplay of nutritional targets, dietary restrictions, taste preferences, and practical considerations like cost or preparation time. At its core, this task involves selecting an optimal subset of food items (e.g., recipes, ingredients) from the vast database of the food knowledge graph. Despite there being a large number of recipes, it essentially boils down to either choosing or rejecting a recipe. Thus, the decision process for each recipe is inherently binary: it is either included in the final recommendation (represented by 1) or excluded (represented by 0). However, the evaluation of a potential recommendation’s quality is quantitative. It depends on how the combination of selected items collectively measures against multiple, often conflicting, real-valued objectives and constraints (e.g., total calories, sodium levels, protein content, overall cost, preference scores). This structure—making binary choices to optimize a quantitative outcome under constraints—positions the problem firmly within the domain of combinatorial optimization.

6.1.1 Rationale for Employing Pseudo-Boolean Optimization

With this motivation, SAT-based Pseudo-Boolean Optimization (PBO) emerges as a highly appropriate mathematical framework for modeling and solving the personalized food recommendation problem. The following sections first introduce the foundational concepts of Pseudo-Boolean Optimization and its integration with SAT-solving techniques, before presenting the rationale for employing this approach within the context of dietary recommendation systems. A boolean, as widely known, is the selection of a ‘true’ or ‘false’ value, or a 0-1 combination. As explained in *“Boolean Functions: Theory, Algorithms, and Applications”* by Crama Hammer (2011) [23], a pseudo-Boolean function is a real-valued function of a finite number of 0–1 variables that maps from :

$$B^n = \{0, 1\}^n \rightarrow \mathbb{R} \quad (\text{Ch. 13, 564}) \quad (6.1)$$

Since PBO inherently handles binary variables, as it concerns functions of inputs restricted to 0 or 1 (Crama Hammer, 2011, Ch. 13), it directly models the discrete selection/non-selection nature of choosing food items. However, while PBO pro-

vides an expressive language to model such complex problems, the task of solving the resulting optimization problem efficiently remains a significant challenge. Optimizing a general pseudo-Boolean function is known to be NP-hard, even for relatively simple quadratic forms (Crama Hammer, 2011, Ch. 13). For practical applications like food recommendation, which may involve hundreds or thousands of potential items and numerous intricate constraints, relying solely on general PBO algorithms might be computationally infeasible. This inherent complexity necessitates leveraging connections to more established optimization paradigms for which highly effective solvers have been developed.

The PBO lays a strong foundation of the food recommendation system’s optimisation as it overlays a connection to well-established solvable frameworks of the 0-1 Integer Linear Programming, as well as the Maximum Satisfiability (MAX-SAT). It gives us the basics of the food recommendation optimization as we know that it shows us the relationship between several recipes and some ‘real value’ that gives us more information about the PBO. This real value is the penalty incurred when a

recipe's constraint is violated— leading to a penalty instead of an instant rejection. With this motivation, we further understand the importance of 0-1 Integer Linear Programming and MAX-SAT and how they are relevant to the food recommendation system. The 0-1 Integer Linear Programming as defined by Integer Programming, MIT [24] is defined as :

$$\text{Maximize } \sum_{j=1}^n c_j x_j \quad (6.2)$$

$$\text{subject to: } \sum_{j=1}^n a_{ij} x_j = b_i \quad (i = 1, 2, \dots, m) \quad (6.3)$$

$$x_j \geq 0 \quad (j = 1, 2, \dots, n) \quad (6.4)$$

$$x_j \text{ integer} \quad (\text{for some or all } j = 1, 2, \dots, n) \quad (6.5)$$

In relation to the recipe recommender, the 0-1 ILP is particularly useful because we minimize the penalty incurred, subject to the linear constraints posed by each recipe condition. That is, for every

$$\sum ax(i) = b(i), \quad \text{where } a \in \{0, 1\} \text{ and } ci \in \{\text{set of constraints}\}, \text{ we incur a penalty } c(j) \text{ that is min} \quad (6.6)$$

Here, it is to be noted that the *maximize* function in the definition can heuristically be interchangeable with *minimize*. Take the following example,

$$\exists \text{ recipes in the FKG : } x(i), i \in [1, n]$$

$$f(x) \oplus c(i), \quad \text{where } c \text{ is some constraint and } \oplus : \{=, >, <, \geq, \leq\} \quad (6.7)$$

then the 0-1 ILB is defined as:

$$\sum ax(i) \oplus c(i) \approx p \quad (\text{penalty} \in \mathbb{R}) \quad (6.8)$$

For a complete 0-1 ILP, the next step is to minimize the penalty created by linear programming the constraints. In my model I have taken the help of Google OR-tools module “minimize”. Therefore, our 0-1 ILP on a high-level description looks something like :

$$(\sum ax(i)) \oplus c(i) \Leftrightarrow p(i) \Leftrightarrow \text{model.Minimize}(\text{sum}(p(i))), \text{ where } 1 < i < n \quad (6.9)$$

Therefore, the variables founded on pseudo boolean optimisation give us a penalty for every recipe that is chosen or not. These penalties are then minimized on the principles of 0-1 integer linear programming, while keeping the constraints error margin as minimum as possible. While PBO and 0-1 ILP explains the penalty minimization for each constraint, the final recommendation system needs to optimize the choice selection of each recipe given all the constraints. This problem is explained with the help of Maximum-Satisfiability or the MAX-SAT problem. The MAX-SAT, as described by Phil Sung, MIT [25] as considering not an arbitrary Boolean expressions but only expressions in conjunctive normal form (CNF), i.e. of the form

$$((A_{11} \vee A_{12} \vee \cdots) \wedge (A_{21} \vee A_{22} \vee \cdots) \wedge \cdots) \quad (6.10)$$

where each literal A_{ij} is either a single variable or its negation, and each clause does not contain more than one literal associated with a single variable. It is important to note that *“in the MAX-SAT variation of SAT, we do not require that all the disjunctive clauses be satisfied. Instead, we want to find an assignment which maximizes the number of satisfied clauses”* (Sung, MIT). The logical structure $((A_{11} \vee A_{12} \vee \cdots) \wedge (A_{21} \vee A_{22} \vee \cdots) \wedge \cdots)$ models a conjunction of disjunctive clauses, where each clause represents a specific condition or constraint.

models a conjunction of disjunctive clauses, where each clause represents a specific dietary or medical constraint. Within each clause, literals such as A_{11} and A_{12} correspond to binary decision variables, indicating whether a recipe is selected (1) or not (0). The downward propagation of variable assignments captures the selection process, while the upward satisfaction of clauses ensures that if any selected recipe satisfies a constraint, the corresponding clause is fulfilled. In this setup, the MAX-SAT objective is to maximize the number of satisfied clauses, leading to a recipe selection that meets as many nutritional and health requirements as possible. This structure effectively connects discrete recipe selection with constraint satisfaction, supporting the personalized recommendation goals of the system. Pseudo-Boolean Optimization provides a rigorous and expressive mathematical framework ideally suited for the complexities of personalized food recommendation. Its ability to handle binary decision variables, quantitative real-valued objectives, complex interactions, and constraints aligns directly with the problem’s nature. The strong theoretical and practical connections between PBO, MAX-SAT, and particularly 0-1 Integer Linear Programming enable the formulation of the recommendation task in

a way that is amenable to solution by powerful, established optimization solvers. Therefore, adopting a PBO approach, often implemented via transformation to and solution as a 0-1 ILP, represents a sound and effective methodology for developing sophisticated food recommendation systems capable of meeting diverse user requirements.

6.1.2 NCD-Based Model Optimisation

This section uses the theoretical grounds of pseudo-boolean optimisation, and applies it to our system through OR Tools’ models. This aims to serve as the bridge between the abstract and the practical. The following equations are derived on the principles of the NCD constraints described in Section 5.5

6.1.2.1 Hypertension

The DASH diet recommends a sodium intake of 1500–2300 mg per 2000 kcal for individuals with hypertension, alongside a potassium-rich diet. The American Heart Association (AHA) advises 3500–5000 mg of potassium daily to help manage blood pressure. Evidence also suggests a minimum dietary fibre intake of 28g/day for women and 38 g/day for men, with each additional 5 g/day reducing systolic and diastolic BP by approximately 2.8 and 2.1 mm Hg, respectively. MedlinePlus further recommends limiting total fat to 27% and saturated fat to 6% of daily caloric intake.

$$model.Add(min_s - current_sodium \leq sodium_under) \quad (6.11)$$

$$model.Add(current_sodium - max_s \leq sodium_over) \quad (6.12)$$

$$model.Add(min_pot - current_potassium \leq pot_under) \quad (6.13)$$

$$model.Add(current_potassium - max_pot \leq pot_over) \quad (6.14)$$

$$model.Add(fibre_min - current_fibre \leq fibre_under) \quad (6.15)$$

$$model.Add(current_satfat - satfat_cap \leq satfat_over) \quad (6.16)$$

$$model.Add(current_fat - fat_cap \leq fat_over) \quad (6.17)$$

6.1.2.2 Diabetes Mellitus

The primary dietary goal in diabetes management is near-normal blood glucose regulation through nutrient-dense, minimally processed foods. Recommendations include low-GI carbohydrates, fiber-rich intake (14 g/1000 kcal), protein at 15–20% of calories (adjusted for kidney function), unsaturated fats, and sodium 2300mg/day. Macronutrient ratios should be individualized.

Glycemic index (GI) and glycemic load (GL) guide carbohydrate quality and insulin response. While Type 1 diabetes requires insulin matched to GL—often clinically impractical—Type 2 focuses on long-term healthy eating, weight control, and avoidance of processed foods and added sugars. Specific caloric targets are to be clinician-guided and are not prescribed in this thesis.

$$model.Add(current_sugar - sugar_cap \leq sugar_over) \quad (6.18)$$

$$model.Add(fibre_min - current_fibre \leq fibre_under) \quad (6.19)$$

$$model.Add(current_satfat - satfat_cap \leq satfat_over) \quad (6.20)$$

$$model.Add(current_sodium - max_s \leq sodium_over) \quad (6.21)$$

6.1.2.3 PCOS

Polycystic Ovary Syndrome (PCOS), affecting 15–18% of women of reproductive age, is linked to insulin resistance and abdominal obesity, even in non-overweight individuals. Dietary management emphasizes reduced-calorie intake, with fat 30% of total energy, saturated fat ideally 8%, and added sugars 5%. A low-glycemic index diet, frequent small meals (every 4 hours), and fiber intake 14 g/1000 kcal are recommended. Timing meals around peak activity supports metabolic efficiency.

Notably, improvements in reproductive and metabolic outcomes can occur independent of weight loss. Nutritional plans should be individualized for long-term adherence and effectiveness.

$$model.Add(slot_cal - cap \leq cal_over) \quad (6.22)$$

$$model.Add(other_slot_cal - active_cal \leq active_shortfall) \quad (6.23)$$

$$model.Add(cal_for("dinner") - int(0.10 \times kcal) \leq dinner_over) \quad (6.24)$$

$$model.Add(current_fat - fat_cap \leq fat_over) \quad (6.25)$$

$$model.Add(current_satfat - sf_cap \leq satfat_over) \quad (6.26)$$

$$model.Add(current_sugar - su_cap \leq sugar_over) \quad (6.27)$$

$$model.Add(min_s - current_sodium \leq sodium_under) \quad (6.28)$$

$$model.Add(current_sodium - max_s \leq sodium_over) \quad (6.29)$$

6.2 Challenges and Mitigation

One of the primary challenges encountered during the development of this project was the creation of a comprehensive ingredient-wise nutritional database. Initially, the process was significantly delayed due to dependency on a collaborative pipeline developed by a partner team, which was experiencing internal bottlenecks. As a result, I was unable to independently proceed with generating a reliable nutritional layer for the ontology. My starting point was the Indian Food Composition Tables (IFCT), which provided nutritional data for only 47 ingredients—insufficient for the level of granularity required in this work. To overcome this limitation, I conducted an extensive search for alternative resources and eventually integrated the

API Ninja Nutritional API into the pipeline. This external API enabled the expansion of the database by contributing nutritional information for an additional 1,398 ingredients. Consequently, the final dataset comprised 1,445 ingredients and 1,021 fully annotated recipes with corresponding nutritional values. This integration marked a critical turning point in the project, enabling accurate nutrient-based constraint modeling. Another key challenge was ensuring adequate recipe variety, especially for users requiring multiple meals per day across several weeks. Although the database currently contains 1,021 nutritionally enriched recipes, this is a subset of the larger scraped dataset comprising 7,177 recipes. The limited number of processed entries was a result of hardware constraints, particularly processing speed and memory availability on the local development system. Despite this, the scalability of the system remains a core strength. The existing pipeline is designed for extensibility and can seamlessly convert additional scraped recipes into structured knowledge base entries. This design choice ensures that future scaling by increasing recipe count can be executed efficiently without architectural changes. Therefore, while the current limitations are tied to computational capacity, the system is well-positioned for rapid expansion once appropriate resources are available. While processing the recipes, I found that many did not calculate their nutritional values correctly. On checking, it became clear that the problem was due to some ingredients missing their nutritional information. Despite efforts to find the exact cause, I was unable to fully understand why certain ingredients failed to process properly. To handle this issue, I added a manual validation step. Recipes with calorie values that seemed much higher or lower than expected were flagged. For these recipes, I manually fetched the missing ingredient information using an external API and corrected the data. Although this approach took additional time, it helped maintain the overall accuracy of the recipe nutrition database.

Chapter 7

Results and Discussions

The current OWL stores 1021 recipes, 969 recipes with nutritional value, 1436 ingredients along with their nutrition, out of which 47 were obtained from INDB and 1398 from API Ninjas. Ontology Processing and Data Representation To support

the recipe recommendation and nutrition analysis system, an ontology-based knowledge base was employed. The ontology, structured in OWL format, captures detailed information regarding food recipes, ingredients, and their nutritional properties using RDF triples. Upon loading the ontology into an RDF graph structure, a total of 72,357 triples were parsed successfully, where each triple represents a factual statement in the form (subject, predicate, object).

- (*Recipe* $\rightarrow$ *hasIngredient* $\rightarrow$ *Ingredient*)
- (*Ingredient* $\rightarrow$ *hasEnergyKcal* $\rightarrow$ *EnergyValue*)

This graph structure enables efficient semantic querying, ingredient resolution, and nutrient property extraction, which are critical for the subsequent stages of recipe recommendation, constraint filtering, and portion management. The loaded on-

tology consists of 72,357 RDF triples, representing detailed relationships between recipes, ingredients, and their nutritional attributes. For instance, a single recipe node such as "Vegetable Pulao" is connected through multiple triples to its constituent ingredients ("rice", "carrot", "peas", etc.), and each ingredient further connects to nutritional properties such as energy (kcal), protein content, and fiber content. On average, each recipe is linked to 5–10 ingredients, and each ingredient

is associated with 5–8 nutrient properties, creating a richly interconnected graph. This dense semantic structure enables the system to perform fine-grained filtering — for example, excluding recipes high in sodium for hypertensive users or prioritizing high-fiber meals for users with PCOS — by reasoning over multiple layers of linked nutritional data. Thus, rather than merely listing recipes, the system dynam-

ically queries and evaluates detailed nutritional and compositional relationships, ensuring that recommendations align closely with user-specific medical, dietary, and lifestyle constraints.

7.1 INDB and API Ninjas

INDB was responsible for 47 ingredients with 39 ingredient properties, whereas API Ninjas gave 1398 ingredients with 10 properties. While INDB dominated the recipe statistics by giving a wider range of nutritional variables, API Ninja outperformed INDB. While contributing with the ten indispensable variables namely Energy (kcal), Carbohydrates (g), Protein (g), Fat (g), Free Sugar (g), Fibre (g), Saturated Fat (g), Sodium (mg), Potassium (mg), Cholesterol (mg), **API Ninjas contributed to 97% of the ingredient nutrition.** The two nutritional databases were processed differently for the ingredients, but ultimately got combined in the recipe’s nutrition calculation. The difference in their property’s integration in the OWL is as shown below :

```
<owl:NamedIndividual rdf:about="#lettuce_leaves">
  <rdf:type rdf:resource="#FoodIngredients"/>
  <hasEnergyKcal rdf:datatype="http://www.w3.org/2001/XMLSchema#string">17.0</hasEnergyKcal>
  <hasCarbGram rdf:datatype="http://www.w3.org/2001/XMLSchema#string">3.3</hasCarbGram>
  <hasProteinGram rdf:datatype="http://www.w3.org/2001/XMLSchema#string">1.2</hasProteinGram>
  <hasFatGram rdf:datatype="http://www.w3.org/2001/XMLSchema#string">0.3</hasFatGram>
  <hasFreeSugarGram rdf:datatype="http://www.w3.org/2001/XMLSchema#string">1.2</hasFreeSugarGram>
  <hasFibreGram rdf:datatype="http://www.w3.org/2001/XMLSchema#string">2.1</hasFibreGram>
  <hasSFAMGram rdf:datatype="http://www.w3.org/2001/XMLSchema#string">0.0</hasSFAMGram>
  <hasSodiumMgGram rdf:datatype="http://www.w3.org/2001/XMLSchema#string">7</hasSodiumMgGram>
  <hasPotassiumMgGram rdf:datatype="http://www.w3.org/2001/XMLSchema#string">30</hasPotassiumMgGram>
  <hasCholesterolMgGram rdf:datatype="http://www.w3.org/2001/XMLSchema#string">0</hasCholesterolMgGram>
</owl:NamedIndividual>
```

Figure 7.1: Ingredient Nutrition by INDB

Stage	Recipes Remaining	Recipes Rejected
All recipes	988	0
Diet filter (non-veg)	988	0
Diet filter (veg)	949	39
Diet filter (vegan)	511	477
Allergy filter (dairy, nonveg)	574	414
Allergy filter (gluten, nonveg)	820	168
Avoid (sandwich) during (breakfast)	974	14
Avoid (dal) during (lunch)	947	41

* Fixed Variables: Age - 23, Sex - F, Activity Level - Active, Weight - 65 kg, Height - 165 cm

Figure 7.4: Sequential Impact of Hard Constraint Filters on Recipe Pool Size

7.3 Soft Constraints – Sample Output: Diabetes

To evaluate the effectiveness of the soft constraint optimization model, we consider a user profile characterized by the following parameters:

- Age: 23 years
- Sex: Female
- Activity Level: Active
- Diet: Non-vegetarian
- Medical Conditions: Diabetes
- Allergies: None
- Food Preferences: Avoid sandwich during breakfast
- Caloric Goal: 2337 kcal

The recipe pool available for each meal time slot was as follows: 338 recipes for breakfast, 602 for lunch, 895 for snacks, and 6 for dinner. The solver returned an

OPTIMAL solution with an objective value of 0.0, indicating perfect satisfaction of all soft constraints under the defined penalty structure.

```
User profile: {'age': 23, 'sex': 'F', 'activity_level': 'active', 'diet': 'non vegetarian', 'allergies': [], 'medical_conditions': {'diabetes': True, 'hypertension': False, 'pcos': False}, 'avoid_food_preference': {'br
eakfast': ['sandwich']}, 'calorie_goal': 2337}

Meal time counts: Counter({'snacks': 895, 'lunch': 602, 'breakfast': 338, 'dinner': 6})
Breakfast recipes available: 338
Lunch recipes available: 602
Snacks recipes available: 895
Dinner recipes available: 6

SOLVER OBJECTIVE VALUE: 0.0
Status: OPTIMAL

--- NCD Wise Nutritional Summary ---

Diabetes in Focus:
Free Sugar (g): Achieved = 116 g, Recommended Target = <= 29 g
Fibre (g): Achieved = 15 g, Recommended Target = >= 32 g
Saturated Fat (g): Achieved = 18 g, Recommended Target = <= 25 g
Sodium (mg): Achieved = 3973 g, Recommended Target = <= 2687 g

Here's your personalized meal plan:

Breakfast:
• Oats Porridge Recipe Oatmeal Porridge (Free Sugar (g): 99 g, Fibre (g): 10 g, Saturated Fat (g): 2 g, Sodium (mg): 85 mg)

Lunch:
• How To Cook Perfect Plain Rice (Free Sugar (g): 0 g, Fibre (g): 1 g, Saturated Fat (g): 0 g, Sodium (mg): 3 mg)

Snacks:
• Easy Aamras Recipe Keri No Ras Homemade Mango Puree Dip Gujarati Aamras Recipe (Free Sugar (g): 15 g, Fibre (g): 4 g, Saturated Fat (g): 0 g, Sodium (mg): 14 mg)

Dinner:
• Masala Egg Fry (Free Sugar (g): 3 g, Fibre (g): 2 g, Saturated Fat (g): 16 g, Sodium (mg): 3872 mg)
```

Figure 7.5: The Recommended Diet for Hypertensive User

Nutrient Summary – Diabetes Focused Targets:

- Free Sugar (g): Achieved = 116 g, Recommended = ≤ 29 g
- Fibre (g): Achieved = 15 g, Recommended = ≥ 32 g
- Saturated Fat (g): Achieved = 18 g, Recommended = ≤ 25 g
- Sodium (mg): Achieved = 3973 mg, Recommended = ≤ 2687 mg

These values may look far from ideal, but this is the best possible outcome given the available recipes. In practice, it's not always possible to meet every nutrition goal exactly — especially when working with real food and real constraints. Not every recipe is designed to fit strict medical targets, so the system finds the closest match it can. The gaps show the real-world challenge of nutrition planning, not a failure of the system.

Meal Plan Recommendation:

- **Breakfast:** *Oats Porridge Recipe Oatmeal Porridge*
Free Sugar: 99 g, Fibre: 10 g, Saturated Fat: 2 g, Sodium: 85 mg
- **Lunch:** *How To Cook Perfect Plain Rice*
Free Sugar: 0 g, Fibre: 1 g, Saturated Fat: 0 g, Sodium: 3 mg

- **Snacks:** *Easy Aamras Recipe Keri No Ras Homemade Mango Puree Dip Gujarati Aamras Recipe*
Free Sugar: 15 g, Fibre: 4 g, Saturated Fat: 0 g, Sodium: 14 mg
- **Dinner:** *Masala Egg Fry*
Free Sugar: 3 g, Fibre: 2 g, Saturated Fat: 16 g, Sodium: 3872 mg

While the solver identified an optimal combination of meals under the caloric and meal timing constraints, it is evident that the soft constraints for diabetes were not fully satisfied. Specifically, the free sugar and sodium levels far exceed the recommended thresholds, and fibre intake falls short. These deviations can inform penalty tuning in future iterations and underscore the need for a broader dinner recipe base to better accommodate diabetic targets.

7.4 Signed Nutrient Gaps: Multi-NCD Performance Discussion

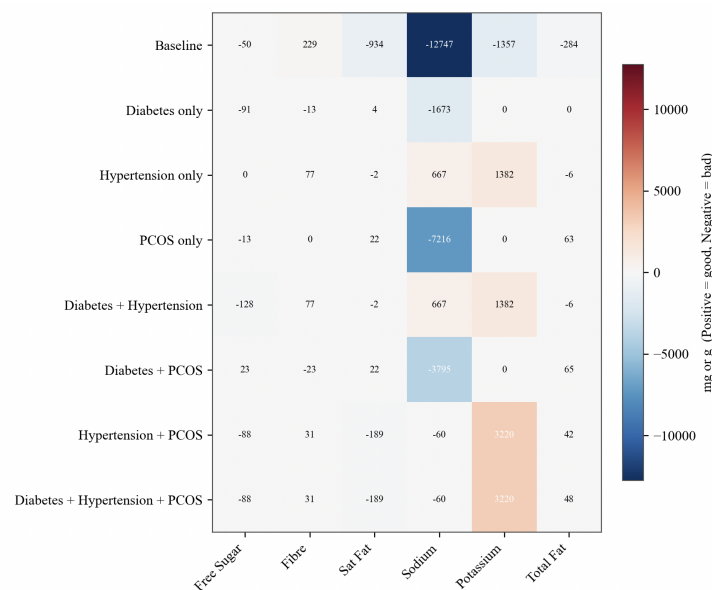


Figure 7.6: Nutrient Gap Matrix Across Disease Profiles

Figure 7.6 shows how nutrient values behave across different NCD profiles. Each cell gives the difference between what was achieved and what was recommended—blue

means the model was below the limit or in control, red means it overshot. It gives a quick snapshot of where the model performed well, and where it fell short.

Looking at the baseline row (no medical conditions), we can see what an unfiltered selection of recipes looks like. Sodium is massively overshot, total fat is high, and fibre is still somewhat reasonable. This isn't surprising—recipes in the wild don't naturally follow clinical guidelines. Once we start layering conditions like diabetes or hypertension, some gaps start shrinking. For example, sodium becomes much more controlled when hypertension is present. The PCOS rows also look cleaner in terms of fat and sugar.

That said, some trade-offs are unavoidable. When trying to balance multiple conditions—like in the Diabetes + Hypertension or Diabetes + PCOS profiles—we can see one nutrient improving while another slips. This is a real outcome of working with finite recipe options: the model is doing its best to juggle multiple constraints, but it's not always possible to meet every target without compromise. Especially when recipes are drawn from real Indian cooking, which might have high-sodium or sugar components baked into cultural norms.

So while this chart might look a bit patchy at first, it actually reflects what happens in real-world planning: sometimes you hit the mark, sometimes you come close, and sometimes you're forced to settle for what's available. Rather than aiming for perfect nutrient scores, the model prioritizes the conditions flagged as most important and tries to balance the rest. Future work could definitely explore smarter substitutions or more flexible constraint handling.

7.5 Conclusion

This thesis presented the design and implementation of a personalized dietary recommendation engine, integrating user-specific medical conditions, dietary preferences, and nutrient goals into a unified optimization framework. What began as a theoretical formulation using constraint satisfaction has been translated into a functioning, context-aware model capable of generating real-world meal plans. At the heart of this system lies the CP-SAT solver from Google OR-Tools, which formulates the meal-planning problem as a Pseudo-Boolean optimization task. By distinguishing between hard constraints—such as allergies, diet type, and ingredient avoidances—and soft constraints—such as ideal nutrient ranges for conditions like diabetes, hypertension, and PCOS—the system ensures feasibility while minimizing deviations through penalty variables.

A key contribution of this work is the integration of an OWL-based nutritional ontology grounded in Indian cuisine, supporting semantic reasoning over food properties. The model dynamically adjusts to varied user profiles, offering meal suggestions tailored by nutrient needs, meal timing, and portioning logic. Importantly, the system acknowledges the practical challenges of dietary planning and focuses on generating realistic, implementable plans that align with health goals. Through successive filtering and optimization, it demonstrates how computational techniques can translate dietary guidelines into tangible, user-specific outputs.

Looking ahead, the framework offers several avenues for real-world deployment and further enhancement. Its modular design and reliance on constraint programming make it well-suited for scaling across clinical, institutional, or public health settings. Applications could range from aiding dietitians and hospitals in individual meal planning to powering mobile apps that support patients in managing chronic conditions through diet.

This thesis shows that optimization is not merely a mathematical tool, but a practical method for navigating the trade-offs between clinical nutrition guidelines and everyday eating habits. The model doesn't promise a perfect diet every time, but it does offer the best possible answer, given who you are and what you need to eat today. And in healthcare and for our well-being, that kind of personalization is exactly what we need more of.

Chapter 8

Extensions and Future Work

8.1 Limitations

A central objective of this thesis was to develop a personalized dietary recommendation system capable of supporting individuals managing non-communicable diseases (NCDs), with a particular emphasis on condition-specific nutritional constraints. In this context, a notable limitation of the current system is the absence of glycemic index (GI) and glycemic load (GL) values within the nutritional modeling. These measures are particularly critical for diabetes management, as they provide insights into how carbohydrate-containing foods affect postprandial blood glucose levels. While glycemic load can, in principle, be computed from existing carbohydrate and fiber data, the glycemic index requires clinical validation and is not readily available for many ingredients—especially those specific to Indian cuisine. The

availability of ingredient-level GI data remains sparse and inconsistent, with most publicly accessible databases offering limited coverage and poor representation of regional Indian food items. As such, incorporating GI values from a small and unrepresentative dataset would have risked introducing bias and reducing the reliability of recommendations for diabetic users. Addressing this limitation—through the integration of a comprehensive, validated GI database—would significantly enhance the system’s ability to model diabetes-specific constraints more accurately and to deliver more nuanced, clinically informed dietary guidance.

8.2 Future Work

This thesis aimed to create a personalized, constraint-driven dietary recommendation system that supports individuals with non-communicable diseases through structured recipe selection and nutritional optimization. While the current architecture establishes a robust foundation, several extensions could enhance the system’s adaptability, accuracy, and inclusivity in future iterations.

8.2.1 User Feedback Integration

One of the core goals of the system is to align recommendations with individual preferences and long-term adherence. Incorporating a feedback loop where users can rate, accept, or reject recipes would enable the system to learn from user behavior over time. This adaptive layer would not only refine the accuracy of recommendations but also increase user satisfaction by dynamically aligning with evolving tastes and compliance patterns.

8.2.2 Enhanced Nutritional Accuracy

The present model calculates nutrition based on raw ingredient values. However, real-world nutrient availability is influenced by cooking methods and nutrient retention factors, which are currently unaccounted for. Integrating nutrient retention data and adjusting nutrition profiles based on preparation methods would significantly improve the system’s precision and real-world applicability—especially important for users managing micronutrient deficiencies or chronic conditions like diabetes.

8.2.3 Expanded Dietary Diversity

The current system is optimized for vegan, vegetarian, and Jain dietary categories, reflecting common constraints within the Indian context. However, expanding the system to support other dietary styles—such as pescetarian or omnivorous diets—would broaden its applicability and make it useful for a wider user base. This would also enable more diverse protein sources and allow better alignment with personalized nutrition goals across different cultural and physiological contexts.

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